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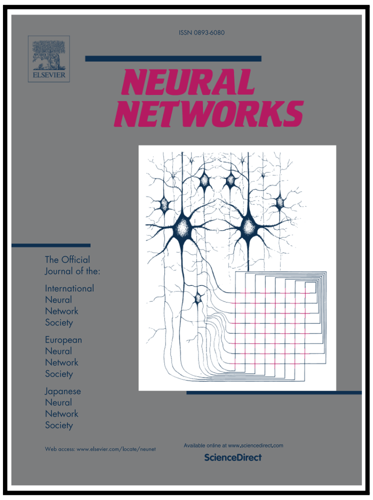

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Reservoir computing with neural activity inputs predicts behavior and neural dynamics in mouse decision-making

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Abstract (187 words)

A major challenge in neuroscience is to elucidate how the brain processes sensory input to generate behavior, especially given the difficulty of measuring neural activity across the whole brain. To address this limitation, previous studies have used artificial neural networks (ANNs) and modeled decision-making circuits in the brain. Here, we used recorded neural activity from mice as additional inputs to a reservoir computer (RC), together with task-related inputs, enabling the RC to operate in a biologically grounded context. We refer to the RC with this hybrid-input condition as a hybrid-input RC (HRC), and to the RC without neural activity input as an artificial-input RC (ARC). Using spike inputs from cortical or subcortical regions, the HRC predicted body movements of head-fixed mice during a task better than the ARC did. This improvement arose not only from the added mouse neuronal activity, but also from the activity generated within the HRC. The generated activity potentially included unrecorded neuronal activity from mice, which was difficult for the ARC. We suggest that the HRC might generate activity patterns resembling unrecorded neural activity, which may contribute to the prediction of animal movements.

Keywords: hybrid neural networks, reservoir computing, biologically constrained artificial intelligence, mouse decision-making, behavior prediction

Introduction

A fundamental function of the brain is making decisions based on sensory inputs. These decisions often accompany the physical body movements of an organism on a continuous time scale[1]. To understand the neural circuit involved in this decision-making process, previous studies have used brain-wide neural recording with two-photon microscopy[2–6] or silicon probes[7–10] to identify the neural correlates of essential task features, such as sensory inputs, action selection, and internal variables of the brain, such as reward expectations[7,11]. However, there are limitations to recording neuronal activity in the whole brain simultaneously, making it difficult to understand how the brain drives body movements for decisions.

In parallel with physiological studies, artificial neural networks (ANNs), including recurrent neural networks (RNNs), reinforcement learning models, and long short-term memory networks, have become successful tools for modeling how the neural circuits of the entire brain drive decision making[1,12,13]. ANNs can perform different strategies[13,14] and often model simplified actions to understand the computational theory of decisions[15]. In addition to focusing on discretized choices, recent studies have used ANNs to investigate how the brain generates physical body movements on a continuous time scale to make decisions[1,16]. These studies showed that the activity of artificial units and that of neurons are somewhat correlated[2,4,12,17], suggesting that the network model provides a hypothesis of how the brain moves the body on a continuous time scale.

Can we develop a brain-like network model in which the artificial units of ANNs exhibit activity patterns similar to those of the real neurons of animals? Although ANNs have become important tools for modeling brain networks[18], limitations have also been reported[19]. Here, we used recorded neural activity from mice as additional inputs to a reservoir computer (RC), which we term a hybrid-input reservoir computer (HRC), enabling the RC to operate in a biologically grounded context. Although previous studies have used either biological signals or artificial task-related signals as the inputs to ANNs[20,21], we adopt both types of the signals for the RC framework for predicting mouse behavior and neural activity. Specifically, we used the HRC to predict the body movements of mice on a continuous time scale, based on both sensory stimuli during a decision-making task and simultaneously recorded neural activity. The performance of HRC was compared with that of a baseline RC trained with only task-related inputs, without neural activity (hereafter referred to as artificial-input RC; ARC).

We chose RC, which only trains readout-layer weights and does not train the internal layers, to test the idea of the HRC in a simple and interpretable model focused on neural activity propagation. In the HRC, neural activity recorded from mice was input into the reservoir and influenced the internal dynamics of artificial units through recurrent propagation. We used the dataset from our previous study in head-fixed mice[22], in which we used a Neuropixels 1.0 probe to electrophysiologically record neural activity from the orbitofrontal cortex (OFC), motor cortex (M1), parietal cortex (PPC), auditory cortex (AC), striatum (STR), and hippocampus (HPC) during a tone-frequency discrimination task.

We found that the HRC outperformed conventional averaging models (AM), generalized linear model (GLM) and the ARC in predicting mouse body movements. This improvement was achieved by spike inputs from almost all the recorded brain regions, except the HPC. We found that the HRC, similar to real mice, predicted choice-dependent body movements even before

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the mice chose the left or right spout. These early choice-dependent movements were not reproduced by the ARC. Further, when we developed an ARC and employed neural activity separately in parallel for prediction, we were unable to match the predictive accuracy of the HRC, indicating that inputting neural signals into the HRC contributed to generating artificial unit activity that enhanced prediction performance. We also trained the HRC using only a subset of the recorded neurons as inputs, holding out the rest. The artificial units in the HRC had activity more similar to the hold-out neurons than did either the ARC units or the subset of neurons actually used as input. Similarly, when we trained the HRC without sound- or choice-relevant neurons, the artificial units were still active at the sound or choice onset. These results suggest that the HRC generates activity patterns resembling neural activity while predicting animal body movements by propagating spike inputs through a recurrent network.

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Results

Hybrid-input reservoir computer (HRC) predicts mouse body movements from task inputs and spike activity.

Our HRC used recorded neural activity from mice, as well as the task parameters, as inputs to a reservoir (RC). To implement the HRC, we used the data from our previous study, in which neuronal activity was electrophysiologically recorded with Neuropixels 1.0 during a tone-frequency discrimination task (**Fig. 1a, b**)[22]. In the task, the mice were head fixed and placed on a cylinder treadmill. In each trial, a sequence of low- or high-frequency pure tones of tone cloud was presented[23–25]. The reward site was either a left or right spout depending on the dominant frequency of the tone cloud (**Methods**). The difficulty of the task varied depending on the proportion of low- and high-frequency tones in the tone cloud. We recorded spiking neural activity from the OFC, M1, PPC, AC, STR, and HPC. The spikes of the simultaneously recorded neurons in each brain region were converted into normalized firing rates (**Methods**) as the spike input and used to train the HRC. Each HRC was trained using inputs from one or two brain regions in addition to the task parameters, whereas the ARC was trained using only task parameters (**Fig. 1c**). Hyperparameters for each of the HRC and ARC were determined separately for each session, and the model was trained on a session-by-session basis. The same session-wise procedure was applied to the other models.

During the task, the body movements of the mice were captured by 4 or 5 video cameras with sampling rates of 137 and 140 Hz in 53 and 35 sessions, respectively (**Fig. 1b**). We extracted the XY trajectories of 30 or 40 body points in the 4- or 5-camera settings with DeepLabCut (DLC)[26]. Using the data collected during the task, the HRC generated the predicted XY trajectories of these body points on a continuous time scale based on the inputs of task parameters (i.e., sounds, rewards, and spout movements) and simultaneously recorded spike data. We trained the HRC using the initial 80% of the data from each session. The RC units were randomly connected based on a set of given hyperparameters. The hyperparameters of RC (i.e., number of nodes, expected number of connections per node and leaking ratio) were selected to achieve the optimal performance with the training data (**Fig. S1**). We then tested the performance of the HRC with the remaining 20% of the data (**Methods**).

We first analyzed the XY trajectories of body points extracted with DLC (**Fig. 1d and S2**). To standardize the frames per second (fps) across all the sessions, the sessions with 140 Hz were downsampled to 137 Hz via linear interpolation. We found that the body trajectories of the mice depended on the sound category (low or high), choice (left or right), and outcome (reward or error). The body movements generated by the HRC varied depending on the task parameters. Using the body movements of the mice and the movements generated by the HRC, we decoded the choices and sound categories via sparse logistic regression with cross-validation (**Fig. 1e**) (**Methods**). In mice, body movements predicted the left or right choice in each trial, and the performance of sound decoding was worse than that of choice decoding. The choice decoding performance in the HRC exceeded the chance level of 0.5 starting at 0.8 s before the choice. To evaluate the contribution of neural inputs, we used an ARC. We also decoded the choice and sound categories using body trajectories generated by the ARC. For sound decoding, the ARC showed higher performance than the HRC and the mice, likely because the ARC was not influenced by neural states and could directly reflect the task sound inputs. The choice decoding accuracy of the ARC began to improve only at 0.3 s before the choice, but exceeded that of the HRC at and after the choice onset. This is presumably for the same reason as above: the ARC receives only the task parameters and is therefore not affected by the variability of the neural inputs, whereas in the HRC the task parameters are mixed with spike input in the reservoir dynamics. In contrast to the ARC, which receives only task parameters and has difficulty predicting the mice's incorrect choices in advance, the HRC receives real neural activity as input, allowing it to generate choice-dependent body movements similar to those of the mice.

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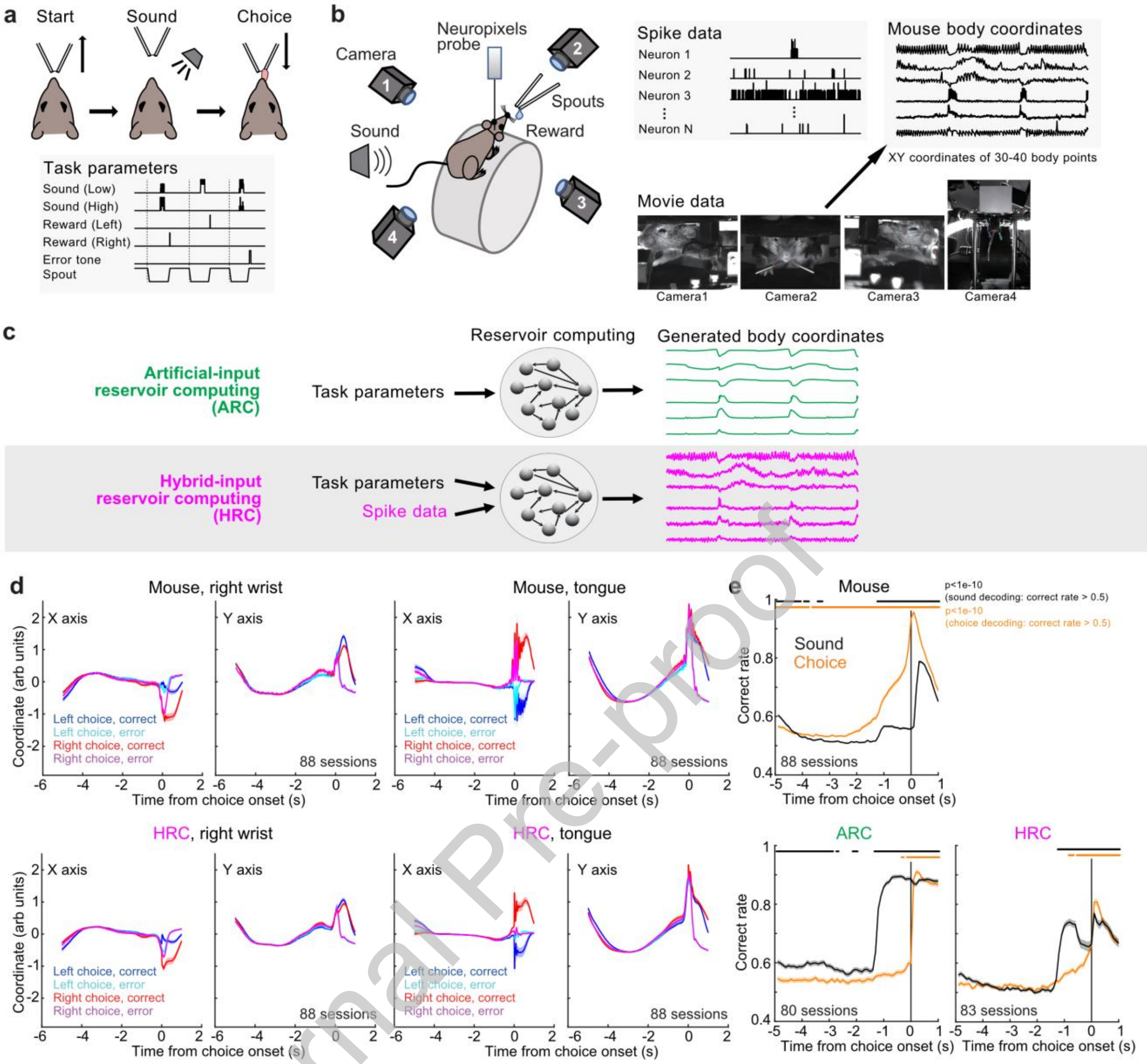

Figure 1. Reservoir computing with and without neural activity inputs (HRC and ARC)

a. Scheme of the tone-frequency discrimination task in head-fixed mice in our previous study[20]. The trial started by moving the spouts away from the mouse. After a sound presentation, the spouts were presented to the mice, and the mice licked the left or right spout to receive a reward. Six task inputs (sound (low), sound (high), reward (left), reward (right), error tone, and spout) were saved as task parameters.

b. Task setting with 4 video cameras and Neuropixels probe. Video data were converted into body-coordinate trajectories using DeepLabCut. Spike activity was simultaneously recorded using Neuropixels probe.

c. Schematic of the ARC and HRC. Task parameters from each session were used as inputs in the ARC, whereas both task parameters and spike data were used as inputs in the HRC to predict body movements.

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(Legend of Figure1)

d. XY trajectories of body points in mice and outputs of the HRC. The average trajectories of the right wrist and tongue are shown (means and standard errors, 88 sessions from 6 mice). **e**. Choice and sound category decoding from body trajectories. The choices and sounds were decoded every 100 ms. The Wilcoxon signed-rank test was used to investigate whether the decoding performance with cross-validation was greater than 0.5 (means and 95% confidence intervals, bars at the top indicate $p < 1e-10$; 88, 80 and 83 sessions from 6 mice for the Mouse, ARC and HRC graphs, respectively).

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The HRC outperforms both the ARC without spike inputs and other machine learning models in predicting mouse body movements.

To validate the performance of RC architecture, we developed two alternative models to estimate mouse body movements (**Fig. 2a**) (**Methods**). The average model (AM) used the mean trajectory of body coordinates in the training data. The generalized linear model (GLM) estimated the body coordinates with linear combinations of 420 task-parameter kernels (i.e., sound, reward/error, and spout movements). We analyzed the root mean squared error (RMSE) between the generated and actual body coordinates in 88 sessions from 6 mice. For all models, we separated the data into an initial 80% (training set) and a final 20% (test set). The RMSE was defined as the average RMSE of all the XY trajectories of body points (60 or 80 XY trajectories for 4 or 5 camera settings) in the test data. We found that the RMSEs of the ARC and the HRC were smaller than those of the AM and GLM (**Fig. 2b**). These results suggest that the RC architecture is suited for behavior prediction.

We next validated the effect of spike input to the HRC. We compared the performance of the HRC and the ARC (**Fig. 2c, 2d**). The RMSEs of the HRC were smaller than those of the ARC, indicating that the spike inputs improved the estimation of body movements. To test whether the improvement could be explained simply by increased input dimensionality, we evaluated an HRC-artificial-input in which spike inputs were replaced with uniformly distributed random inputs. This model not only failed to outperform the HRC but also performed worse than the ARC (**Fig. 2c**). We further found that the performance of the HRC was better than that of the HRC with 3-second-shifted spike inputs (HRC shift-3s) (**Methods) (Fig. 2c**), suggesting that the improvement in performance was not achieved simply by the presence of spike-like inputs or their spectral properties. Rather, it suggests that the precise timing of spikes is crucial for generating body movements. We also found that the prediction accuracy of the HRC was better than that of the HRC without task-parameter inputs (HRC no env.), indicating that neural activity inputs alone were not sufficient to generate accurate body movements (**Fig. 2c**). These results suggest that the inputs of both neural activity and task parameters are essential for generating mouse body movements over a continuous time scale.

To investigate whether the improvement in prediction accuracy comes from the neural activity itself or from processing the spike inputs as unit activity within the reservoir network, we evaluated a control condition in which the spike inputs were placed in parallel with the RC units—without providing them to the RC (ARC-spike-parallel; ARCSP). The predictive accuracy of ARCSP was higher than that of the ARC, but lower than that of the HRC (**Fig. 2b, 2c**). This result suggests that the prediction performance is further improved by the internal processing of spike inputs in the HRC, in which the spike inputs and the task parameters are nonlinearly projected into high dimensions and their temporal history is accumulated in the reservoir state.

Next, we investigated whether the performance of the HRC depended on the brain regions of the recorded neurons. We observed improvements in the HRC performance in almost all the recorded brain regions except the HPC (Wilcoxon signed rank test: AC, PPC, STR, M1, and OFC, $p = 0.030$–$6.2\text{e-}6$; HPC $p = 0.080$) (**Fig. 2d)**. We also examined whether the performance of the HRC depended on the number of recorded neurons. To normalize the variability in RMSE across sessions, we used ΔRMSE ratios defined as the percentage change in RMSE upon adding spike inputs, computed as RMSE of the HRC relative to that of the ARC. Negative values indicate $\text{RMSE\_HRC} < \text{RMSE\_ARC}$ (i.e., improved performance with spike inputs), whereas positive values indicate worse performance (**Methods**). We found that the ΔRMSE ratios were correlated with the number of neurons (**Fig. 2e**). When analyzed by brain region, the number of neurons in the PPC and OFC showed significant correlations (**Fig. 2e**). To explore this further, we selected sessions containing data from at least 40 neurons and investigated how the RMSE changed when we randomly selected 1, 5, 10, 20, or 40 neurons from the same neuron group (**Fig. 2f**). This method reduced the variabilities in HRC performance caused by the sessions (i.e., differences in the number of recorded neurons and mouse body movements). We found that the HRC performance improved as the number of spike inputs increased, particularly for inputs from the PPC (**Fig. 2f**). We then directly compared the ΔRMSE ratios across brain regions and found that using PPC neurons yielded slightly better predictions of body movements than neurons from other regions (**Fig. 2g**). Among these comparisons, using 20 PPC neurons yielded significantly better prediction than using 20 AC neurons (**Fig. 2g**). These results suggest that the PPC had distributed representations of body movements across neurons.

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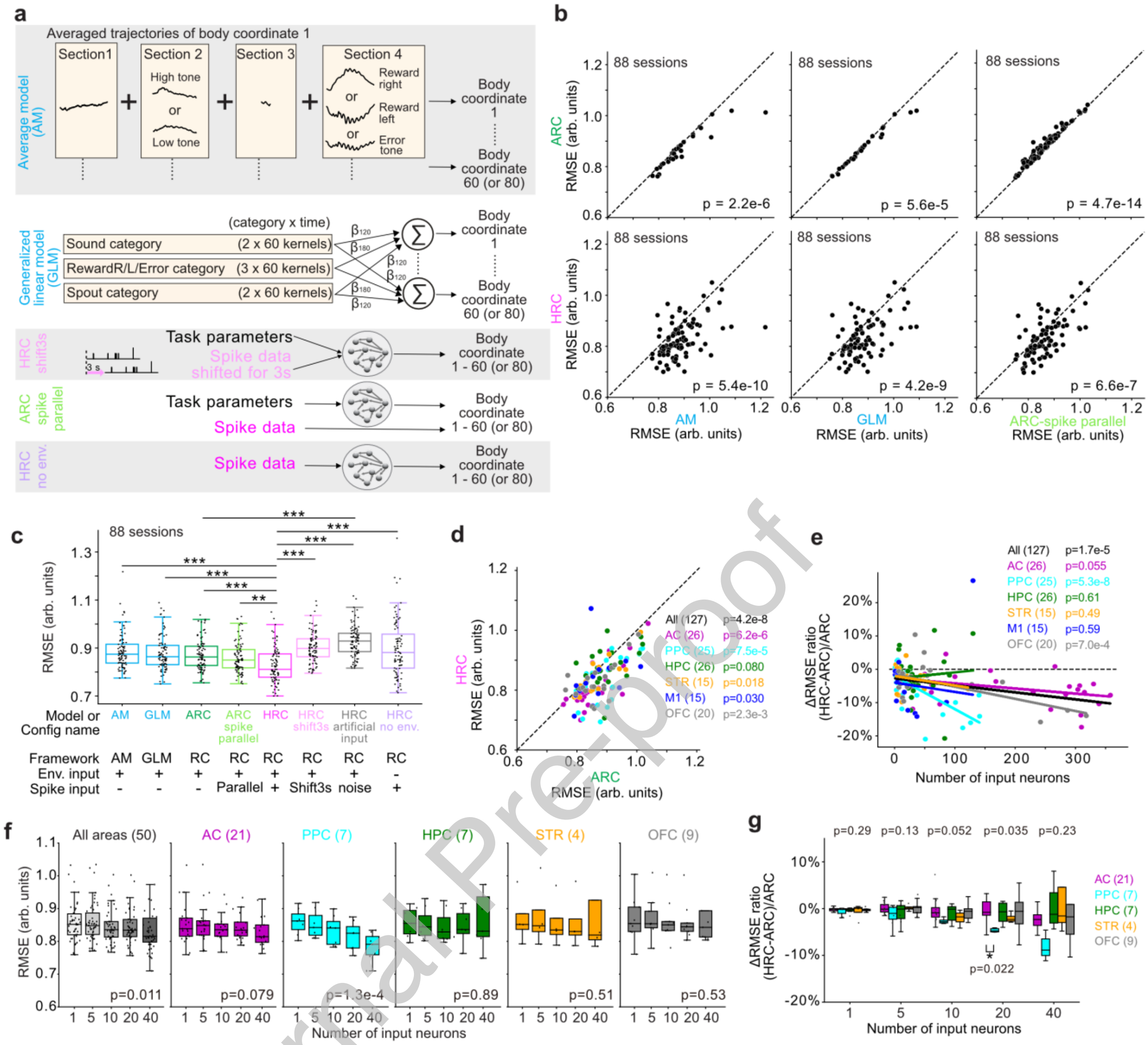

Figure 2. Performance of the hybrid-input reservoir computer (HRC)

a. Scheme of the average model (AM), generalized linear model (GLM), HRC shift3s, ARC–spike parallel (ARCSP), and HRC-no-env. The AM calculates the mean trajectories of each body point in the training data (the first 80% of all the data) divided into four sections: (1) before tone onset, (2) between tone onset and spout arrival, (3) between spout arrival and choice, and (4) after choice. We compared the body trajectories in the test data (the last 20% of all the data) with the mean trajectories in the training data to compute the root mean squared error (RMSE) at each time point. The GLM used 60 kernels for each task category (**Methods**). Sound kernels started from tone onset (section 2) and lasted 6 s with steps of 0.1 s (60 kernels). The reward and error kernels started from the onset of the outcome and lasted 6 s (section 4). Spout kernels started at the arrival and departure of spouts and lasted 6 s (sections 1 and 3). We trained the GLM with the first 80% of all the data and tested it on the last 20% of all the data. HRC shift3s used spike inputs that were temporally shifted by 3 s to disrupt the alignment between neural activity and behavior. ARC–spike parallel (ARCSP) used spike activity as an additional predictor for behavior without providing it to the reservoir. HRC-no-env used only the spike inputs without task parameters.

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b. RMSEs of the HRC and ARC in each session compared with those of the AM, GLM, and ARC-spike parallel (ARCSP) (Wilcoxon signed-rank test in 88 sessions from 6 mice).
c. Summary of HRC performance (**: $p < 1e\text{-}5$, ***: $p < 1e\text{-}7$ in Wilcoxon signed-rank test, 88 sessions from 6 mice). The central line and edges in the boxplots represent the median, 25th, and 75th percentiles, respectively; the whiskers represent most extreme data points inside the limits of 1.5 times the interquartile range, here and throughout the paper.
d. Improvements in performance in HRC with precise spike inputs. RMSEs of RC with and without spike inputs (HRC and ARC). The neural activity from the PPC-HPC or STR-M1 was simultaneously recorded with one Neuropixels probe in 24 or 15 sessions, respectively (39 sessions in total out of 88 sessions). The performance of the HRC using spike inputs from each brain region was independently analyzed (plots show the data from 88 + 39 = 127 sessions in total from 6 mice) (**Methods**) (Wilcoxon signed-rank test).
e. Relationship between the number of spike inputs and the performance of the HRC. The ΔRMSE ratio was the relative change in the RMSE of the HRC compared with that of the ARC (**Methods**). Robust linear regression was used to analyze the significance of the slope (t test).
f. Relationship between the number of neurons and the performance of the HRC within sessions. Sessions with 40 neurons or more were used. A total of 1, 5, 10, 20, or 40 neurons were randomly chosen from each session. Since only 2 sessions had data from M1, we did not show the individual plots of M1. Robust linear regression was used to analyze the significance of the slope (p values from the t test). Outliers with RMSE > 1.1 were excluded from the figures. For **f** and **g**, plots including all the data points are shown in **Fig. S3**. Boxplot definitions are as in **c**.
g. Comparisons of ΔRMSE ratios across brain regions for each number of neurons (p values at the top: one-way ANOVA; p values at the bottom: Tukey's HSD post hoc tests). Boxplot definitions are as in **c**. Outliers with a Δ RMSE ratio > 10% were excluded from the figures.

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Compared with the artificial units in the ARC, those in the HRC exhibit more neuron-like activity.

To investigate how the HRC achieved better performance than the ARC, we first examined the output weights that connect each unit to body movement. Because output weights depend on the strength of regularization, we added a stronger regularization condition (ridge = 1e-2) in addition to the original condition (ridge = 1e-7). Even under the stronger regularization condition, the HRC still outperformed the ARC in behavior prediction (**Fig. S3c, left**). We then compared the root mean square (RMS) of the output weights across models. Under the stronger regularization condition, the HRC and ARC units showed comparable output weights (**Fig. S3c, middle**), whereas under the weaker regularization condition, the output weights of the ARC were significantly larger than those of the HRC (**Fig. 3a, left**). RMS corresponds to the L2 norm of the weights and can be interpreted as an index of the magnitude of the solution or the required parameter strength. This result suggests that the output mapping of the ARC was less stable under weak regularization, whereas that of the HRC was more stable. We also calculated the Gini coefficient to examine the unevenness of output weights across units and found that the HRC consistently showed lower values than the ARC regardless of regularization strength, indicating that the distributed representation in the HRC was robust to the strength of regularization (**Fig. 3a, right, S3c, right**).

We analyzed the internal dynamics of the artificial units of the reservoir (**Fig. 3b-d**). Compared with the ARC, in which dynamics were driven only by low-dimensional task inputs, the HRC received higher-dimensional neural inputs that were further transformed nonlinearly within the reservoir[27,28]. As a result, the HRC required a larger number of principal components to achieve high cumulative explained variance (**Fig. 3b**). We then used cosine distance to quantify the similarity between the activities of real neurons and those of HRC and ARC units. The results showed that the neuronal activity recorded from mice was more similar to the HRC units than to the ARC units (**Fig. 3c**).

Among all the neurons recorded from the AC, PPC, HPC, STR, M1, and OFC (**Fig. 2e**), we investigated the task-relevant neurons that exhibited increased activity during the task, as in our previous studies[7,22,23] (**Methods**). Specifically, task-relevant neurons showed increased activity in at least one time window (0.1 s) between -1.5 s and 2.5 s from the sound onset (40 windows) or between -0.5 s and 2.5 s from the choice (30 windows, 40 + 30 = 70 windows in total) compared with the baseline activity ($p < 1e-10$ in one-sided Wilcoxon signed-rank test). The baseline activity was recorded from -0.2 s to 0 s from spout removal and -0.2 s to 0 s from spout approach for the sound- and choice-aligned activities, respectively. Using the same criteria, we identified the task-relevant artificial units in the HRC and ARC based on the activity of the RC units. We found that the activity of neurons in mice and that of artificial units in the HRC were aligned with the task events of sounds or choices (**Fig. 3d**). We analyzed the proportion of task-relevant neurons activated at sound and choice onsets and verified that the activity of HRC units was more closely related to the task events than that of the ARC (**Fig. 3e**).

In addition to analyzing the temporal patterns of artificial units, we analyzed the representations of real neurons and artificial units by introducing the tone index (**Methods**)[7,23]. The tone index was used to compare the activity during the low- and high-category tone trials and ranged between -1 and 1. We independently analyzed the tone indices for correct and incorrect trials based on the neural activity before the sound, during the sound, and after the choice. When neurons had significantly different activities during correct low- and high-category tone trials (p < 0.01 in the Mann–Whitney U test) and the tone indices of correct and incorrect trials had the same signs, we defined the neurons as sound-responsive. When the signs of the tone indices of correct and incorrect trials were opposite, we defined the neurons as choice responsive.

We analyzed the tone indices before sound according to the tone categories in previous trials (**Fig. 3f**). We found that the mouse neurons and the HRC units had distributed tone indices, whereas the ARC units had the same signs of tone indices between the correct and incorrect trials, suggesting that the ARC units represented previous sound categories, which was inconsistent with the neurons in mice. We compared the ratio of sound-responsive to choice-responsive neurons in three different time windows in each session (**Fig. 3g**). We found that a greater proportion of artificial units in the HRC were choice-responsive than the ARC before sound and after choice, which is consistent with the findings in neurons in mice. Because neural activity itself is highly variable, the HRC units also exhibited variable activity. However, this variability was not merely a consequence of increased variance; the HRC units also showed temporal features and task-related representations that were more similar to neural activity than those of the ARC units.

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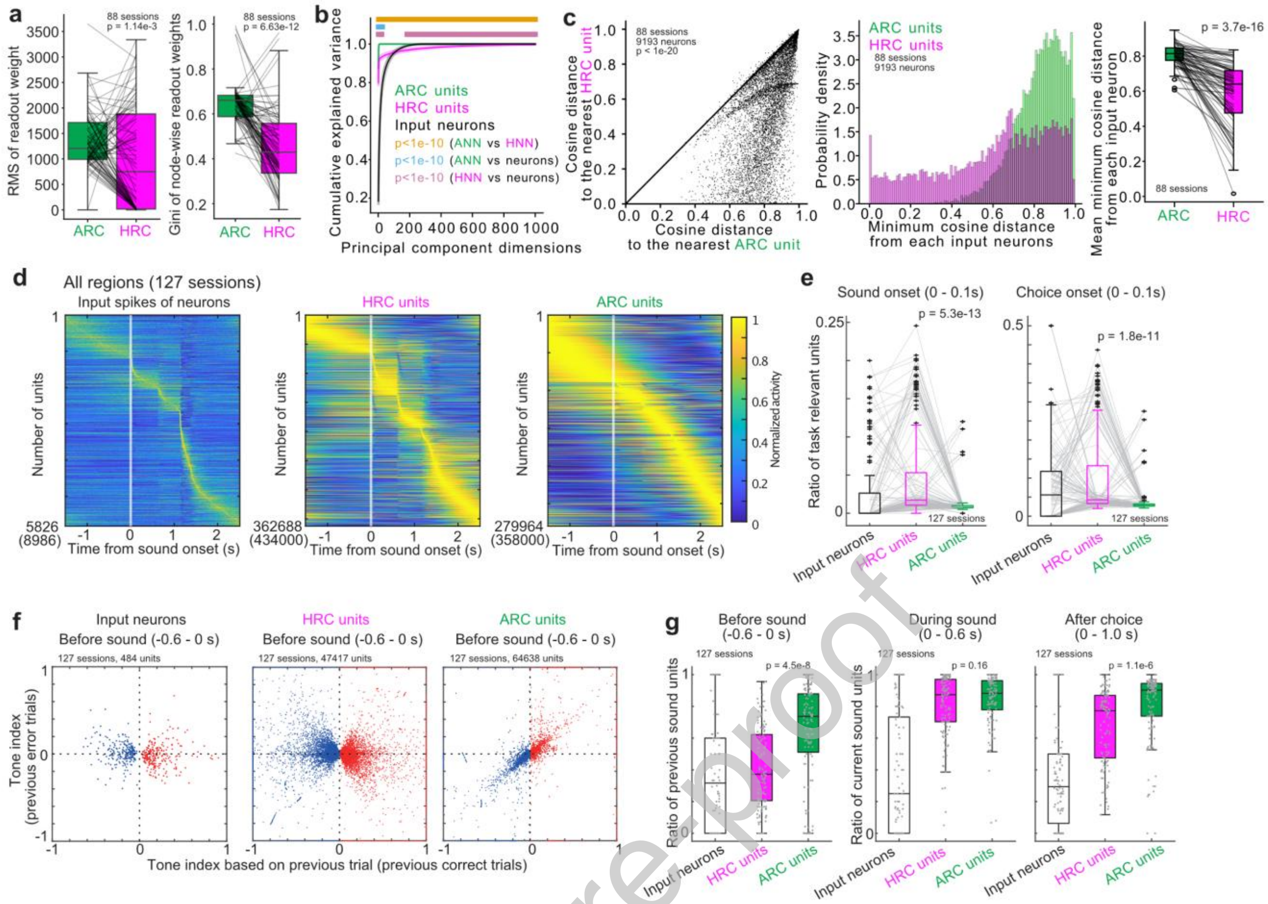

Figure 3. The activity of artificial units in the HRC is similar to that of neurons in mice.
a. Weight in HRC and ARC. RMS of readout weight of the HRC with ridge=1e-7 (**left**). Gini of node-wise readout weight of the HRC with ridge=1e-7 (**right**). Boxplot definitions are as in **Fig. 2c**. (Wilcoxon signed-rank test in 88 sessions from 6 mice)
b. Cumulative explained variance from principal component analysis for the ARC, HRC units, and input neurons. Means and standard errors of the mean (88 sessions from 6 mice for the ARC, HRC units and input neurons). When the number of principal components exceeded the number of input neurons (median 51.5), we set the explained variance to 1. For the PC dimensions beyond 184, the explained variance of input neurons was consistently lower than that of the HRC units (Wilcoxon signed-rank test).

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c. Cosine distances from each mouse neural activity to the nearest ARC or HRC unit activity. Each neuron's cosine distance to its nearest HRC and ARC unit was scatter-plotted (**left**). The distribution of minimum cosine distances is shown as a histogram (**center**). For each session, the average of the minimum cosine distance was calculated for the ARC and the HRC units (**right**). The activity of neurons in mice was significantly closer to that of units in the HRC than to that of the ARC (Wilcoxon signed-rank test). Boxplot definitions are as in **a**.
d. Average activity of task-relevant units with increased activity during the task. The spike input of neurons is presented as the average activity of mouse neurons across trials from the AC, PPC, HPC, STR, M1, and OFC. The HRC and the ARC units are presented as the average activities of the artificial units. Parentheses show the number of all units across sessions. The order of task-relevant units was sorted according to the maximum activity timings. The average activity of each unit was standardized between 0 and 1.
e. Proportion of task-relevant units at sound and choice onsets in each session. Compared with those of the ARC, the activities of the artificial units in the HRC were aligned with the task events (Wilcoxon signed-rank test). Boxplot definitions are as in **a**.
f. Tone indices of mouse neurons and artificial units before sound. We analyzed the task-relevant units with increased activity before the sound (-0.6–0 s) and significant differences in activity between the previous correct trials with low- and high-category tones ($p < 0.01$ in the Mann–Whitney U test). The tone indices of the mouse neurons and the HRC units were widely distributed.
g. Ratio between the sound- and choice-responsive units. The sound- and choice-responsive units had tone indices with the same and opposite signs in the correct and incorrect trials, respectively. Similar to **c**, we analyzed the task-relevant units with increased activity at the corresponding time windows and different activities between the low- and high-category correct trials. The ratio of sound-responsive neurons was smaller in the HRC units than in the RC units before sound and after choice (Wilcoxon signed-rank test). Boxplot definitions are as in **a**.

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The population trajectories of artificial units in HRC exhibit choice modulations before choice onset, which is consistent with mouse neurons.

Decoding analyses using the generated body movements from the HRC and the ARC revealed that the HRC achieved faster choice separations than the ARC, consistent with the findings in mice (**Fig. 1e**). Single-unit analyses further indicated that the artificial units of the HRC represented the choices (**Fig. 3**). To further investigate choice representations in the HRC, we investigated the population trajectories of the HRC units and compared them with those of mouse neurons and ARC units (**Fig. S4**). For this analysis, we used 97 sessions out of 127 sessions in which at least 20 neurons were recorded in each brain region. We found that the population decoding trajectories were diverged between left and right choices at -0.1–0 s in the mouse neurons, the HRC units and the ARC units (Wilcoxon signed-rank test: p = 4.0e-3–4.3e-12) (**Fig. S4a-c**). Direct comparisons of choice separations revealed that the HRC units had significantly greater separations than the mouse neurons and the ARC units did (**Fig. S4d, e**).

These results suggest that the HRC generates choice-relevant unit activity by combining internal-state information contained in partially recorded neural activity with task parameters, and by propagating and amplifying this information. Together with the result that the behavioral prediction accuracy in the ARCSP was lower than that in the HRC (**Fig. 2b**), this indicates that neural-activity inputs were integrated with task parameters and processed within the HRC, thereby generating additional information relevant to behavior.

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Artificial units in the HRC generate activity patterns resembling unrecorded neural activity.

A potential advantage of the HRC is that the artificial units can resemble the neural activity of unrecorded neurons for predicting how the whole brain generates body movements for decision-making. To test this hypothesis, we first examined whether the HRC could generate the activity patterns corresponding to unrecorded neurons by quantifying the similarity between unit activity and neuronal activity using cosine distance.

We selected mouse neurons that were close to the ARC units based on cosine distance, defined as near neurons, and constructed a "near-HRC" using only the near neurons as inputs to the reservoir. We then compared the cosine distances across four categories: (1) the HRC units, (2) the ARC units, (3) near neurons, and (4) far neurons that were not used as the input to near-HRC. We found that the activity of the near-HRC units was more similar to that of the far neurons than that of the ARC units, both at the single-neuron level (**Fig. 4a**) and across sessions (**Fig. 4b**). These results suggest that the HRC generated unit activity that was similar to the mouse neuronal activity, compared to the ARC, regardless of whether those neurons were used as the input to the reservoir. Moreover, compared with the near neurons which were provided to the HRC, the near-HRC units were closer to the far neurons (**Fig. 4b**), suggesting that the HRC did not simply reflect its neural inputs but could generate activity resembling unrecorded neurons through nonlinear propagation within the network[27,28].

To examine whether this effect differed across brain regions, we analyzed sessions in which multiple brain regions were recorded simultaneously and measured the distance between HRC units trained with input from one brain region and neural activity in the other brain region (**Fig. 4c**). We did not detect a significant difference in this analysis.

To further investigate whether the HRC generated activity similar to the activity of unrecorded neurons, we examined the activation timing of the HRC units around sound onset. We held-out the task-relevant neurons that exhibited increased activity during sound presentation (0.6 s) (defined as sound neurons) from the input to the HRC and trained the HRC without these neurons (**Fig. 4c**). In total, we held-out 803 out of 5389 task-relevant neurons, which were recorded in 73 out of 127 sessions. We then compared the activity of the HRC units with and without sound neuron inputs across these 73 sessions (**Fig. 4d**). Even in the absence of input from the sound neurons, the artificial units in the HRC remained active at sound and choice onset. Similarly, we defined choice neurons which exhibited increased activity during choice (0.6 s). When 1,898 choice neurons were held-out from the input, the HRC units also remained active at sound and choice onset. Furthermore, similar results were obtained when 803 or 1,898 randomly selected neurons were held-out (**Fig. S5**). These results suggest that the HRC generates neuron-like activity patterns resembling unrecorded neural activity by propagating spike inputs through artificial units in recurrent neural network dynamics.

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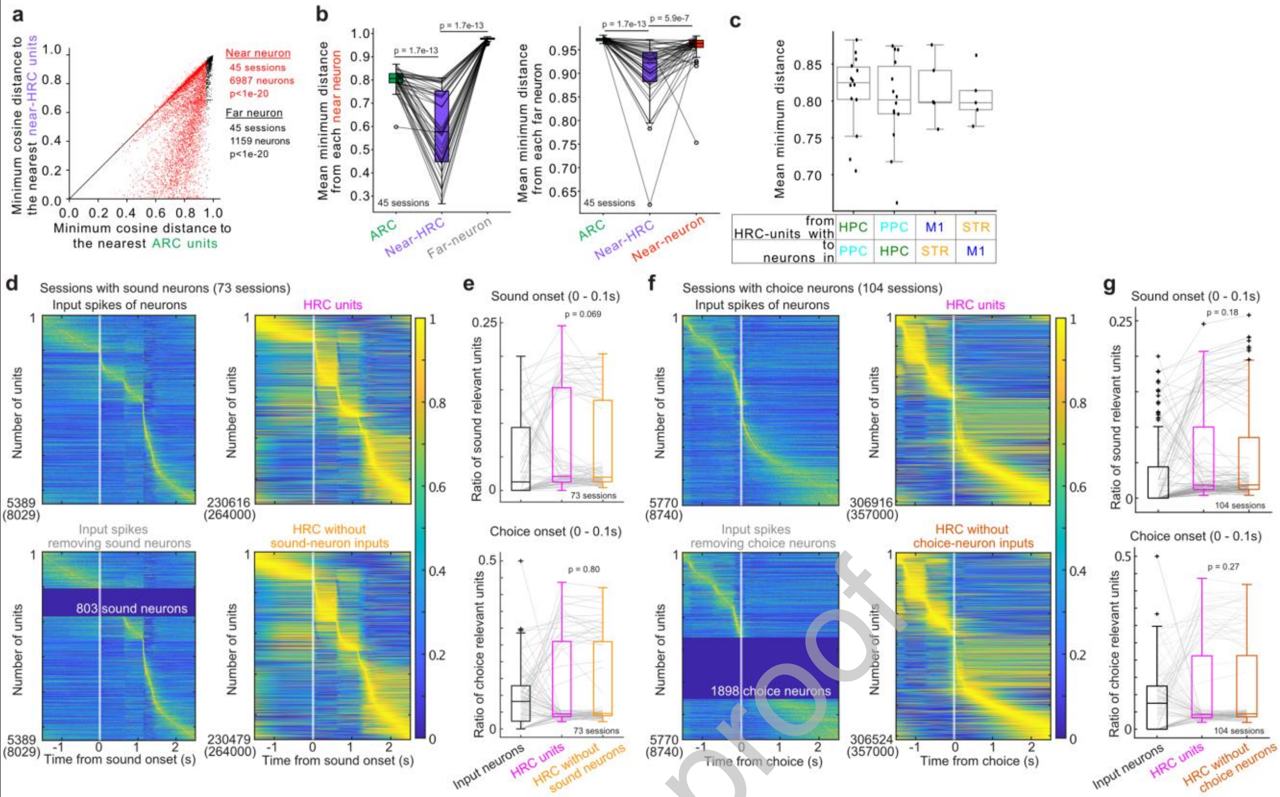

Figure 4. Artificial units in the HRC exhibit activity patterns similar to unrecorded neural activity of mice.

a. Cosine distances from each mouse neural activity to the nearest ARC or HRC unit activity. Input neurons were categorized as far or near neurons according to the cosine distance from the ARC units (threshold = 0.95). We selected 45 sessions which had at least five far neurons to avoid bias from single-neuron variabilities. The activity of mouse neurons was significantly more similar to that of the HRC units than to that of the ARC units (Wilcoxon signed-rank test).

b. Mean of the minimum cosine distance between each neuron (far or near neuron) and its nearest ARC unit, HRC unit, or other neuron in each session. The minimum cosine distance was calculated for each neuron and then averaged across neurons within each session. The mean cosine distances of both near and far neurons were smaller for the near-HRC units than for the ARC units or other neurons (Wilcoxon signed-rank test with Bonferroni correction). Boxplot definitions are as in **Fig. 2d**.

c. Cosine distances between HRC units with input from one brain region and neural activity from another brain region in sessions in which multiple brain regions were recorded. No significant differences were detected among region combinations. Boxplot definitions are as in **Fig. 2c**. (Mann–Whitney U test with Bonferroni correction; all $p > 0.05$)

d. Average activity of task-relevant units with increased activity during the task. The data plot definitions are as in **Fig. 3d**, but we used the 73 sessions in which the sound neurons were recorded. HRCs using spike inputs with (top) and without (bottom) sound neurons are shown.

(Figure legend continues in next page)

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(Legend of Figure 4)

e. Proportion of task-relevant units at sound and choice onsets in each session. The spike inputs with or without the sound neurons did not significantly affect the proportion of sound- and choice-onset activity of artificial units in the HRC (Wilcoxon signed-rank test). The data plot definitions are as in **Fig. 3e** but from 73 sessions.
f. Same visualization as in panel **d**, but performed for 104 sessions containing choice-responsive neurons and temporally aligned to choice onset.
g. Proportion of task-relevant units at sound and choice onsets in each session. The spike inputs with or without the choice neurons did not significantly affect the proportion of sound- and choice-onset activity of artificial units in the HRC (Wilcoxon signed-rank test). The data plot definitions are as in **Fig. 3e** but from 104 sessions.

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Discussion

In this study, we implemented a hybrid-input reservoir computer (HRC) that uses both task parameters and the activity of real neurons in mice as inputs for reservoir computer (RC) to predict the body movements of mice. The sensory inputs of sounds, rewards, and spout movements were presented to the mice during the tone-frequency discrimination task in our previous study[22].

We found that the HRC outperformed an artificial-input reservoir computer (ARC) that used only task parameters and not neural activity as input (**Fig. 2**). We also found that performance deteriorated when uniformly distributed artificial inputs were provided instead of spike inputs, and that the HRC with 3-s-shifted spike inputs performed worse than the original HRC. These results indicate that temporally aligned spike activity is important for performance improvement. The RC with spike inputs alone (HRC-no-env.) did not outperform the ARC, suggesting that the recorded spikes did not capture all the sensory inputs. The HRC was independently implemented using neurons from one or two brain regions in each session. To capture all the sensory inputs, further experiments that simultaneously record 1) the neural activity across multiple brain regions and 2) a large population of neurons with a wide-field two-photon microscope[29–31] or multiple Neuropixels probes[32] are needed.

Decoding analyses using behavior generated by the HRC and ARC showed that choice decoding rose significantly earlier in the HRC than in the ARC (**Fig. 1e**). In contrast, the period during which sound decoding was significantly elevated was shorter in the HRC than in the real mice, whereas it was longer in the ARC. Real mice can make incorrect left/right choices because their decisions depend on internal states as well as sensory evidence, whereas the ARC, which does not receive neural activity as input, can reflect task parameters more directly. This may explain why sound decoding was higher in the ARC than in the mice. The lower sound decoding in the HRC compared with the mice may indicate that neural inputs exerted a stronger influence than task parameters in some conditions. More effective ways of combining task parameters and neural activity for input to RC may therefore further improve the prediction.

We found that the performance of the HRC was improved by spike inputs from almost all the recorded regions except the HPC (**Fig. 2d**). As previous studies have shown that neural activity is modulated by body and facial movements[33–36], it was not surprising that the neurons in many brain regions contributed to generating movements on a continuous time scale. The HPC may not be directly involved in controlling body movements, but is instead considered to play a role in higher-order aspects of behavioral control, such as action planning, sequence prediction, and prospective coding of future behavior[37–39]. We also found that the accuracy of the HRC was improved by increasing the number of spike inputs from the PPC (**Fig. 2f**). The PPC not only represents choices in perceptual decision making[40] but also changes activity during inter-trial intervals[41] and estimates stimulus probabilities[42], expected outcomes[43,44], and hidden contexts[45]. Assuming that such diverse internal representations are distributed across multiple neurons, the accuracy of body movement prediction may increase in a neuron number-dependent manner. Although the primary motor cortex (M1) is thought to be the center of behavior output and its improvement was statistically significant, the effect was relatively moderate compared to other areas. Previous studies suggest that the M1 is deeply involved in action initiation, controlling movement direction and force parameters, and adaptive fine-tuning. On the other hand, there is also evidence indicating that M1 does not faithfully map the spatial coordinates of each body part[46–48]. Thus, for a task like ours that requires predicting the coordinates of body parts, relying on M1 activity may not have yielded sufficient performance.

Next, we found that the activity of the HRC units was more similar to spike activity than that of the ARC (**Fig. 3**). In addition, the HRC outperformed the ARCSP in behavior prediction (**Fig. 2b**). PCA showed that the HRC representations were higher-dimensional than the original neuronal activity and ARC representation (**Fig. 3b**). Consistent with this interpretation, hyperparameter tuning showed that the optimal number of reservoir nodes was smaller for the ARC than for the HRC (**Fig. S1d**). This result suggests that when only task parameters are provided as inputs to the reservoir, a relatively small number of nodes is sufficient for predicting body movements, and increasing the number of nodes does not effectively contribute to the representation. In contrast, when neural activity is provided as additional input, the reservoir can utilize a larger number of nodes to generate higher-dimensional unit dynamics. This expansion of internal representations may contribute to the higher variance observed in HRC units and ultimately improve behavioral prediction performance. The high variability of the HRC units likely arose from the interaction between neural inputs and recurrent reservoir dynamics. Because individual units integrated different combinations of neural and task inputs over multiple time scales, the HRC activity became more heterogeneous and less redundant. Furthermore, population decoding analysis showed that the HRC achieved better choice separation than both the ARC and real neurons (**Fig. S4**). Given the task structure, in which choices are presumably determined by both sound-related task information and internal-state information, it is natural that the HRC, which receives both types of information, showed better choice separation than the ARC or real neurons. One possible explanation is that the recorded neurons capture only a subset of the neural information in the brain, whereas the task parameters provided to the HRC complement the partially observed neural signals. In addition, the HRC explicitly receives task parameters such as sound inputs, whereas the recorded neurons provide only partial observations of the brain state, which may also contribute to the stronger choice separation observed in the HRC. Taken together, these results suggest that the activity of the HRC units contains additional movement information beyond that present in the original neural activity.

We investigated whether the artificial units of our HRC could predict the neural activity of unrecorded neurons (**Fig. 4**). Even when we developed the HRC without the inputs of neurons with specific activity patterns, the HRC units still exhibited activity patterns that resembled those of the omitted neurons, suggesting that the HRC could generate activity patterns resembling the missing neural activity. In addition, even when sound- or choice-relevant neurons were excluded from the input in separate analysis, the HRC units were activated at the time of sound or choice onset, respectively. Notably, the HRC units accurately predicted behavior with relatively smaller output weights than the ARC under weaker ridge regularization. In other words, the internal unit dynamics of the HRC contributed substantially to predicting body movements. To further investigate the functional role of these internally generated unit activities and whether they generated unit activity similar to neural activity in specific brain areas, future work could utilize simultaneous recordings from multiple brain areas. Such data would allow us to assess whether inputs from one area can generate activity in another area.

We developed the HRC using a simple reservoir architecture in which the internal weights were not updated from their initial random values by assuming that the brain itself utilizes somewhat random connectivity to propagate information[49]. Although this simple reservoir architecture led to notable improvements of behavioral prediction, as shown in this study, it is likely that further enhancements could be achieved by incorporating architectures with trainable internal connections, or fixed but biologically structured internal connections, such as connectome-based reservoirs[50,51].

One advantage of developing brain-like AI is that pseudoinactivation of the HRC may simulate changes in mouse behavior in inactivation experiments using optogenetics or other methods (**Fig. S6**) (**Supplementary Methods**). To test this hypothesis, we trained an HRC using the neural activity of the OFC as the input, as in **Fig. 1**. We then removed the spike inputs from the HRC to represent pseudoinactivation (**Fig. S6**). We found that although optogenetic inhibition of the OFC significantly reduced tone history-dependent choice biases in mice during the task, pseudoinactivation of the HRC only slightly reduced discrete choice biases. This discrepancy may be explained by the relatively small number of recorded OFC neurons used as input to the HRC; pseudo-inactivating only a small subset of OFC neurons may have a limited effect compared to the broad suppression achieved by optogenetics. These findings suggest that accurately replicating the effects of optogenetic inactivation may require recording from a larger population of OFC neurons, or, beyond merely silencing recorded inputs, targeting HRC units that exhibit OFC-like dynamics.

Some parameters were fixed to reduce computational cost, based on pilot analyses showing little qualitative change in the results across a small number of sessions. More exhaustive searches, such as random search[52], may identify better combinations in future work. For the leaking rate, we selected two extreme values, 0.05 and 1, to examine both slow integration dynamics and rapidly updating dynamics. Additional analyses in three sessions using several combinations of fixed hyperparameters (iss, sr, ridge, and lr; **Fig. S1b,c**) suggested that the main conclusions were not substantially altered, although a limitation remains that these values may affect the operating regime of the reservoir dynamics and thereby influence performance in both the ARC and HRC.

In summary, we implemented an HRC that uses both task parameters and neural activity recorded from mice as input to achieve more accurate predictions of mouse body movements than a conventional ARC trained with only task-related artificial input. The activity of the HRC's artificial units showed activity patterns closer to that of recorded neurons than the ARC units, as measured by cosine distance. These units were temporally aligned with task events and reflected the mice's choices—resembling the activity patterns observed in recorded neurons. Notably, simply inputting neural activity into a reservoir computing network—without updating internal weights—enabled the HRC to generate activity patterns resembling those of unrecorded neurons. While further validation using a larger number of spike inputs and alternative architectures will be essential, this HRC offers a promising approach for understanding brain function through a form of data augmentation: propagating observed neural activity through a randomly connected network to extend the representation of neural activity and improve behavior prediction accuracy.

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Methods

We used a hybrid-input reservoir computer (HRC) within a reservoir computing framework, in which the neural activity recorded from mice during a behavioral task was used as input. The animal experiments and datasets used in this study were reported in detail in our previous study[53].

Animal experiments

All animal procedures were approved by the Animal Care and Use Committee at the Institute for Quantitative Biosciences (IQB), The University of Tokyo. We used 14 male CBA/J mice (Charles River, Japan) that were 8 to 15 weeks of age at the start of behavioral training with electrophysiology[53]. We recorded the body movements of 6 of the 14 mice with either 4 or 5 cameras (1 and 5 mice, respectively) placed around each mouse during a task. The data from these 6 mice were used in this study. The mice were housed in a temperature-controlled room with a 12 h/12 h light/dark cycle. Before surgery, 3 mice were housed in one cage. The mice were allowed ad libitum access to food, and their water intake was restricted to 1.5 mL per day. The weights of the mice were checked daily to prevent dehydration. The mice were caged in isolation after craniotomy.

The surgical procedures used were previously described[7,23,53,54]. Our surgery involved two steps: the implantation of a lightweight head bar and a craniotomy for electrophysiology. In each surgery, the mice were anesthetized via an intraperitoneal injection of a mixture of medetomidine (0.3 mg/kg), midazolam (4.0 mg/kg), and butorphanol (5.0 mg/kg) or isoflurane (2% at induction, below 1.5% for maintenance). Meloxicam (2 mg/kg, subcutaneous) and eye ointment were also used. The electrophysiology recording site was chosen from the following candidates: the recording sites of the OFC were centered at +2.6 mm anterior–posterior (AP) and ±1.4 mm medio–lateral (ML) from bregma; the sites for the STR and M1 were +0.8 mm AP and ±1.5 mm ML; the sites for the PPC and HPC were -2.0 mm AP and ±1.7 mm ML; and the sites for the AC were -3.0 mm AP and ± 3.8 mm ML.

- Electrophysiology during a tone-frequency discrimination task

The behavioral task was explained in detail in our previous study[53]. Briefly, the task was based on the tone-frequency discrimination task of previous studies[24,25,55]. The mice were head fixed and positioned over a cylinder treadmill. One precalibrated speaker (#60108, Avisoft Bioacoustics) was placed diagonally to the right of each mouse for auditory stimulation (Type 4939, Brüel and Kjaer). Water was delivered through two spouts connected to an infrared lick sensor (Ohara, Inc.). The behavioral system was controlled by a custom MATLAB (MathWorks) program running on the Bpod r0.5 framework (https://sanworks.io) in Windows.

Each mouse selected the left or right spout depending on the dominant frequency of a tone cloud to receive a water reward[24,55]. The frequency of each sound pulse in the tone cloud was sampled from 18 logarithmically spaced slots (5-40 kHz). The tone cloud in each trial contained low-frequency (5–10 kHz) and high-frequency (20–40 kHz) sounds and was categorized as low or high depending on the dominant frequency. The high- or low-category tone in a correct trial was associated with a reward of 2.4 μl of 10% sucrose water in the left or right spout. An incorrect trial resulted in a noise sound burst for 0.2 s. When the mice did not select a spout within 15 s from the start of the trial, a new trial started.

Our task had two task conditions for sound presentations on the basis of the transition probability 'p' of the tone category in each trial[53]. In the repeating condition, the tone category was repeated often in each trial (p = 0.2), whereas in the alternating condition, the tone category was altered by 80% (p = 0.8). Each mouse was assigned to one of the two task conditions (1 and 5 mice assigned to the repeating and alternating conditions, respectively).

All the electrophysiological recordings were performed with Neuropixels 1.0 (IMEC) inside the sound-attenuating booth (Ohara, Inc.). A subset of neurons from 6 out of 14 mice reported in our previous study were used[53]. Every day, i.e., every session of the behavioral task, we inserted one Neuropixels probe that targeted either the OFC, STR/M1, PPC/HPC or AC. The probe location was identified in the post hoc fixed brain by labeling the probe injection site with CM-DiI or DiA at the recording phase (Thermo Fisher Product #V22888 or #D3883). The Open-Ephys GUI was used to acquire the neural data at a sampling rate of 30 kHz with a gain of 500 (PXIe acquisition module, IMEC)[56]. The task events were sampled at 2.5 kHz (BNC-2110, National Instruments). Spike sorting and manual curation were performed with Kilosort 3 on MATLAB (https://github.com/cortex-lab/KiloSort) and Phy on Python (https://github.com/cortex-lab/phy).

We defined the depth of each unit according to the approximate location of the probe and the electrode position, which measured the average maximum amplitude of spikes. For the OFC recording, the units with an estimated spike depth from the brain surface less than 1.9 mm were analyzed. For the PPC or HPC, the units with an estimated spike depth less than 1.0 mm or between 1 mm and 2.3 mm, respectively, were analyzed. For the STR or M1, the units with an estimated spike depth greater than or less than 1.5 mm, respectively, were analyzed. All the recorded units for the AC probe were analyzed.

As described in our previous studies[7,53], after electrophysiological recording, the mice were deeply anesthetized with isoflurane (5%) and further anesthetized with a mixture of 1.5 mg/kg medetomidine, 20 mg/kg midazolam, and 25 mg/kg butorphanol. The mice were perfused with 10% formalin solution to identify the probe locations on fixed brain slices.

Measurement of mouse body movement

During the tone-frequency discrimination task, we captured the mouse body movement with 5 video cameras in 5 mice across 69 sessions (DMK 33UX273, Imaging Source). For the other mouse, we captured body movements with 4 cameras in the first 8 sessions and 5 cameras for the remaining 11 sessions (88 sessions in total) (ICCapture Express). The front camera was placed in front of the face of the mouse to capture tongue movement. The front left and front right cameras were placed at the side of the mouse's face to capture the forelimbs and whiskers. The other 1 or 2 back cameras, depending on the setting, were placed behind the mouse to capture the trunk, hind limbs, and tail. The frame rate of the video was 137 Hz on average, with a standard deviation of 0.00046 Hz, for the first 53 sessions and 140 Hz, with a standard deviation of 0.00237 Hz, for the last 35 sessions. The frame rates were different both without and with the light-emitting-diode (LED) devices used for video alignment.

From the video data from each camera, we first extracted the XY trajectories of body points with DeepLabCut (DLC)[26]. From the front camera data, we extracted 6 body points, namely, nose, tongue, right foreleg finger, right foreleg wrist, left foreleg finger, and left foreleg wrist. From the front left and front right camera data, we extracted 7 positions each, namely, nose, side upper eye, side bottom eye, right foreleg finger, right foreleg wrist, left foreleg finger, and

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left foreleg wrist. From the back camera data, we extracted 10 positions each, namely, right hindleg toe, right hindleg heel, right hindleg knee, left hindleg toe, left hindleg heel, left hindleg knee, back top, tail tip, tail middle, and tail root. In total, we extracted 30 and 40 body points from the 4- and 5-camera settings, respectively. In addition to the body points, we also extracted the movements of the spouts in each of the videos for temporal alignment.

We used data from several sessions to train the DLC: 10 sessions (2 sessions each from 5 mice) for the front camera, 5 sessions (1 session each from 5 mice) for the front left and front right cameras, 1 session from 1 mouse for the back camera, and 5 sessions (1 session each from 5 mice) for the back left and back right cameras. In each session, we randomly sampled approximately 100 frames from each video and manually labeled the body points. We extracted 996, 500, 500, 100, 500, and 499 frames for the front, front left, front right, back, back left, and back right cameras, respectively. The labeled frames from the same camera were used to train one DLC model, which was applied to all the sessions from the corresponding camera across the mice. After the automatic extraction of body points with DLC, we detected outlier points with a likelihood of less than 0.5. These outliers were replaced with linear interpolations with the nearest points with likelihood above 0.5. The trajectory of each body point was z scored with a mean of 0 and a standard deviation of 1.

For the back cameras, the appearance of spout movements differed across sessions. We thus trained 2 and 3 DLC models for the back and back left cameras, respectively, and applied the proper DLC model for spout movement extraction.

Time alignment of movie data from all the cameras and task parameters

To temporally align the movies from different cameras, we extracted the spout movements in all the movies with DLC in 55 out of 88 sessions in 6 mice. In the other 33 sessions, all the cameras detected a 10-ms infrared light stimulus (660 nm) from the LED in each trial, which was used for time alignment. The LED stimulus was also sampled at 2.5 kHz as the task parameter (BNC-2110, National Instruments).

- Time alignment with spout movements

For each camera, we detected the spout movement by using either the x or y coordinate with a large variance. On the basis of the spout coordinates from all the cameras, we used the *finddelay* function in MATLAB to analyze the differences in movie frames among cameras. We then deleted the initial and final frames in each camera to coarsely align the camera data. The aligned spout coordinates from all the cameras were combined and averaged.

Using the spout movement detected by a rotary encoder (BNC-2110) with a sampling rate of 2.5 kHz and the average spout movement from all the cameras, we determined the frames per second (fps) of the video data. The average fps was 140 Hz (35 sessions) or 137 Hz (53 sessions). We then downsampled the task events that were originally sampled at 2.5 kHz to match the camera fps, ensuring that no task events were lost in the process.

Subsequently, fine discrepancies due to frame drops or other issues in the behavioral data from each camera were corrected on a trial-by-trial basis. For the spout data obtained from the input data and from 4 or 5 cameras, we calculated a moving median over 201 frames (100 frames before and after). In parallel, thresholds were calculated by applying a moving average over 30,001 frames for all of the spout data. The data were then binarized to 0 or 1 depending on

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whether the spout moving medians were above or below each threshold. The frame numbers at which the values of these binarized spout data changed were identified as the spout movement timing frame numbers for each trial. By calculating the difference in spout movement frame numbers between the input data and the camera data for each trial, the error in frame count for the spout movement timing was obtained for each trial.

A moving median over 21 trials (10 trials before and after) was calculated for the trial-by-trial spout movement timing error data. The moving median was used to avoid overcorrecting for trial-specific noise influenced by errors in the automatic labeling by DLC. Finally, the body coordinate data were adjusted by reducing or increasing the number of frames by the amount of this moving-median error frame count, followed by linear interpolation. This linear interpolation was applied to all body coordinate data, including the spout movement data, thereby temporally aligning the coordinate data with the input data.

- Time alignment with LED

In each video, we analyzed the average luminance of a rectangular region including the LED stimulus in each frame. By computing the differences between the luminance values in consecutive frames and setting any negative differences to zero, we created LED data that contained positive values for the frames where the LED was lit. We calculated the moving median for these LED data over a total of 301 frames (150 frames before and after) and subtracted it from the LED data to eliminate luminance changes with periods longer than 300 frames. The corrected LED data were then z scored (i.e., mean 0, std 1). A threshold was manually determined (either 0.3 or 0.2), and the data were binarized to 0 or 1 according to this threshold. In sessions where incorrect threshold crossings were visually confirmed, we set the value to 1 only when the threshold-crossing frame was at least 500 frames after the previous frame, since intervals between trials were always more than 500 frames. The frame numbers at the moments when the value changed from 0 to 1 (LED-on frame numbers) were obtained and used for time alignment.

First, to align the frame numbers of the first LED-on frames across the five camera images, we deleted the initial frames from the body coordinate data of the five cameras so that the first LED-on frames aligned. Next, for each of the body coordinate datasets 1 to 5, we obtained the LED-on frame numbers of the first and last trials and calculated the total number of frames between them, i.e., the total frame count for all trials. Similarly, we obtained the frame numbers corresponding to the trial initiation timings in the input data (which are the same as the LED emission timings) and calculated the total frame count for all trials between the first and last trials.

By comparing the longest total frame count among the 5 body datasets with that of the 1000 Hz input data, we determined the fps of the body data. The input data were then uniformly downsampled by deleting frames at regular intervals to match this fps.

Finally, to ensure that the number of frames per trial was consistent, we performed linear interpolation so that the LED-on timings in body datasets 1 to 5 aligned with the trial initiation timings in the input data for each trial. During interpolation, we recorded the number of frames deleted or added for each trial and manually confirmed that there were no obviously incorrect frame adjustments exceeding 100 frames.

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Data analysis

The models using reservoir computer were programmed with Python (3.11.4 and 3.6.10). Other models and data analysis were performed with MATLAB (MathWorks) and Python (3.11.4 and 3.6.10).

We used the data from 6 out of 14 mice from our previous study[53], as body movements were captured by cameras only in these 6 mice over 88 sessions. The neural data from the PPC-HPC and STR-M1 were simultaneously recorded with one Neuropixels probe. We analyzed the data from 27 sessions for the PPC-HPC (no neurons in the PPC in 2 sessions or the HPC in 1 session), 15 sessions for the STR-M1, 26 sessions for the AC, and 20 sessions for the OFC. Mice u01 – u06 had data from 19, 17, 20, 2, 19, and 11 sessions, respectively. We selected the sessions for analyses using the same criteria used in our previous study[53]: (i) the percentage of correct responses for both the 100% low-tone and 100% high-tone stimuli was greater than 75%, and (ii) the total reward amount in one session was at least 600 µL.

Modeling the body movements of the mice

- Reservoir computer

We used reservoir computer (RC) (ReservoirPy, Python), a type of recurrent neural network[55], to generate the body movements of the mice during the tone-frequency discrimination task. The RC was trained on the weights of the full connections between the recurrent units and outputs via ridge regression. The recurrent units were leaky-integrator neurons with a leaking rate lr of either 0.05 or 1. The connections (i) between the RC inputs and the recurrent units and (ii) among recurrent units were probabilistically determined. Win was randomly sampled from {−1, 0, 1}, whereas W was sampled from a normal distribution. The connection probabilities were treated as hyperparameters and optimized (see below). The activation function of the recurrent units was the hyperbolic tangent (*tanh*) function. The update of states was described with the following equation with x_t representing the current state, x_{t+1} representing the next state, u representing the input vector, W_{in} representing the weight between inputs and RC units, and W representing weights among RC units:

$$x_{t+1} = (1 - lr)x_t + lr * tanh(W_{in} * u + W * x_t) \quad (1)$$

The RC inputs were the short pure-tone pulses in the tone cloud (low or high category), left or right water reward, error tone, and spout movement, which were downsampled to 140 Hz (35 sessions) or 137 Hz (53 sessions) to align with the movie data. The water rewards were defined to deliver between 0 and 20 ms from the reward onset. All the inputs ranged between 0 and 1. There were 6 task inputs in total (**Fig. 1a**).

- Hybrid-input reservoir computer (HRC)

Our HRC used recorded neuronal activity as the RC inputs as well as the 6 task inputs. The neuronal activity was averaged and downsampled to 140 Hz (35 sessions) or 137 Hz (53 sessions). The spike counts in each time window were scaled and ranged between 0 and 1.

In addition to using time-aligned mouse neural activity as the RC inputs, we used time-shifted activity as the input to investigate whether the precise timing of neural activity contributed to generating the body movements of the mice (**Fig. 2c**). We used the neural activity before 3 s as the input to the HRC by putting the last 3-s activity at the initial timing of the session. We also constructed the HRC with spike inputs but without the task inputs.

- *Training and testing of the ARC and HRC*

All models were trained and evaluated separately for each session. In each session, we used the first 80% of the data for training the RC and the last 20% of the data for testing. Within the training data, we optimized the hyperparameters (**Fig. S1**). We tested the 2 leaking rates of each unit (0.05 or 1) and the 44 combinations of average connections per unit as W and number of nodes (for 1 or 3 connections per unit: 1000 to 7000 nodes with every 1000 increments; for 2, 4, or 5 connections: 1000 to 5000 nodes with every 1000 nodes; for 10, 30, or 100 connections: 3000 to 7000 nodes with every 1000 nodes). There were 88 hyperparameters in total. The spectral radius (*sr* = 0.9), input scaling (*iss* = 1), and ridge penalty (*ridge* = 1e-7) were fixed to reduce computational cost, based on pilot analyses showing little qualitative change in the main conclusions across a small number of sessions.

We used the first 80% of the training data (the first 64% of all the data in each session) and trained the RC with each of the 88 hyperparameters. The root mean square error (RMSE) of the RC was determined with the last 20% of the training data (16% of all the data) to identify the hyperparameter setting with the smallest RMSE. The RMSE was defined as the average RMSE of all the XY trajectories of body points (60 or 80 XY trajectories for 4 or 5 camera settings, respectively). Using the identified hyperparameter setting, we trained the RC with all the training data (80% of all the data) and analyzed the performance with the test data (the last 20% of all the data); the test data were used in neither the hyperparameter setting nor the connection weights of the RC.

We defined the ΔRMSE ratio of the HRC on the basis of the RMSE with no spike inputs (i.e., ARC) to evaluate the relative change in performance by adding spike inputs (**Fig. 2e, g, S3b**). Negative values indicate improved performance (smaller RMSE), whereas positive values indicate worse performance:

$$\Delta RMSE\ ratio\ (\%) = \frac{RMSE_{spike} - RMSE_{no\ spike}}{RMSE_{no\ spike}} * 100 \qquad (2)$$

When we analyzed the performance of HRC with 1, 5, 10, 20, or 40 input spikes (**Fig. 2f, g**), we used the hyperparameter settings for which the RC achieved the best median performances in 23 sessions of 1–5 spike inputs (1 and 5 neurons), 27 sessions of 6–15 spike inputs (10 neurons), 20 sessions of 16–29 spike inputs (20 neurons), and 23 sessions of 30–50 spike inputs (40 neurons) (**Fig. S1**). In each session, we randomly selected neurons with spike inputs of 1, 5, 10, 20, and 40 for 10 times and performed HRC to analyze the median RMSE for the 10 iterations.

- *Average model*

We divided each trial into four sections: (1) before tone onset, (2) between tone onset and spout arrival, (3) between spout arrival and choice, and (4) after choice. In each section, we divided the trials into different categories and analyzed the mean trajectories of body points: Section 1 had one category; Section 2 had low- and high-tone categories; Section 3 had one category; and Section 4 had three categories: left reward, right reward and error tone (**Fig. 2a**). The mean trajectories of the body points were analyzed using the first 80% of the data in each session (training data). We then compared the trajectories of the last 20% of the data (test data) with those of the mean trajectory model to analyze the RMSE.

- Generalized linear model

We used the MATLAB software package glmnet (*cvglmnet*) (https://glmnet.stanford.edu/index.html) (**Fig. 2a**)[57]. Our generalized linear model (GLM) included 60 kernels for each task event. The sound kernels (low and high) started from sound onset. The right reward, left reward, and error kernels started from the onset of the outcome. The spout kernels started from the arrival and departure of the spout. Each task-event kernel was 0.1 s and set for 6 s from the start. The initial 80% of the data in each session were used to train the GLM with 10-fold cross-validation (CV), whereas the last 20% of the data (test data) were used to compute the RMSE.

- Decoding sound and choice categories from body movements

We decoded the sound categories (low or high) and the choices of the mice (left or right) from the body movements of the mice or those generated by RC (**Fig. 1e**). We used the MATLAB software package glmnet (https://glmnet.stanford.edu/index.html) with L1 regularization. In each time window of 0.1 s, we averaged the body movements and performed sound and choice decoding. The performance of the decoder was evaluated with 5-fold CV. Due to a shortage of test trials, some folds contained fewer than three trials or only a single class label, making logistic classification ill-defined or numerically unstable in some sessions for the ARC and HRC; in these cases, we used the 3-fold CV. If the 3-fold CV was still ill-defined or unstable, we discarded the session. For decoding from the mouse data, we analyzed all the 88 sessions with 5-fold CV. For ARC, 62, 18, and 8 sessions were 5-fold CV, 3-fold CV, and discarded, respectively. For HRC, 65, 18, and 5 sessions were 5-fold CV, 3-fold CV, and discarded, respectively. To remove the contaminations of sound and choice decoding, we subsampled the trials used to train the decoder such that the number of trials for the 4 sound-choice combinations were identical (i.e., low-left, low-right, high-left, and high-right). We repeated the CVs 10 times with different subsamples and analyzed the average accuracy rates.

Output weight analysis

- Root mean square (RMS) of readout weights

The root mean square (RMS) of the readout weights was computed as

$$RMS = \sqrt{\frac{1}{M}\sum_{k=1}^{M} w_k^2}$$

where w_k denotes each element of the readout weight matrix and M is the total number of weights. The RMS corresponds to the L2 norm of the readout weights normalized by their number and reflects the overall magnitude of the solution.

- Gini coefficient

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The Gini coefficient was computed for each session using the standard pairwise definition based on differences between unit contributions:

$$G = \frac{\sum_i \sum_j |w_i - w_j|}{2N^2\overline{w}}$$

where w_i denotes the mean absolute output weight of unit i, N is the number of units, and $\overline{w}$ is the mean of w_i across units. The RMS was computed across all readout weights to quantify the overall magnitude of the output mapping, whereas the Gini coefficient was computed from node-level contribution magnitudes in order to capture how output information was distributed across units.

Comparison of activity between neurons in mice and artificial units in RC

- Cosine distance

To quantify the similarity between the activity of artificial units and real neurons, we compute the cosine distance between their activity vectors. The pairwise cosine distances were calculated as follows:

$$cosine\ distance = 1 - \frac{activity\ vector1 \cdot activity\ vector2}{\|activity\ vector1\|\ \|activity\ vector2\|} \tag{3}$$

A smaller cosine distance indicates greater similarity in the temporal activity pattern.

- Task relevant neurons and units

We analyzed the activity of neurons in mice and that of artificial units in the HRC and ARC on the basis of our previous studies[7,53]. We first identified the task-relevant neurons and units with increased activity during the task compared with baseline activity (one-sided Wilcoxon signed-rank test, $p < 1e\text{-}10$). Like in our previous study of mouse neurons[53], for sound-aligned activity, task-relevant units presented increased activity in at least one time window (0.1 s) between -1.5 s and 2.5 s from sound onset (40 windows). For choice-aligned activity, task-relevant units exhibited increased activity in at least one time window between -0.5 and 2.5 s from the choice (30 windows, in total 40 + 30 = 70 windows). The baseline for the sound-aligned activity was -0.2 to 0 s from spout removal, i.e., before trial initiation. The baseline for the choice-aligned activity was -0.2 to 0 s from the time the spout approached, i.e., between the end of the sound and making a choice.

Using the average activity across trials, we identified the maximum activity timings of task-relevant units and compared the timings across units in the mice and models (**Fig. 3d**). We then analyzed the tone index of each unit on the basis of the left and right reward-associated tone trials[7,23,53]:

$$Tone\ Index = \frac{mean(spike\ (right\ tone\ trials)) - mean(spike\ (left\ tone\ trials))}{mean(spike\ (right\ tone\ trials)) + mean(spike\ (left\ tone\ trials))} \tag{4}$$

The tone index ranged between -1 and 1 and was independently analyzed for correct and incorrect trials. The tone indices before sound (-0.6–0 s) were analyzed on the basis of correct and incorrect previous trials, whereas the indices during sound (0–0.6 s) and after choice (0–1.0 s) were analyzed for the current trials[53]. On the basis of the tone indices and the differences in activity between the left- and right-choice correct trials ($p < 0.01$ in the Mann–Whitney U test), we analyzed the proportions of sound- and choice-responsive neurons before the sound, during the sound, and after the choice (**Fig. 3e, f**). For the sound- and choice-responsive neuron deletion experiments, we removed from the HRC inputs task-relevant neurons whose activity peaked within 0-600 ms after sound onset or within 0-600 ms after choice, respectively (Fig. 4). In total, 803 sound-responsive neurons and 1,898 choice-responsive neurons were removed. As control experiments, we also removed the same number of randomly selected neurons (Fig. S5).

Population decoding analysis of neurons in mice and artificial units in RC

We used a decoding-based analysis described in recent studies[58,59]. For the RC, we analyzed all the 88 sessions. For the mouse neurons and HRC, we used 97 out of 127 sessions in which more than 20 neurons were simultaneously recorded from each brain region. All the population analyses were performed in each session separately (**Fig. S4**).

We first binned the data into 50 ms between -2.0 s and 1.0 s from the choice onset. We then applied principal component analysis (PCA) to reduce the data to 20 principal components. We used a linear support vector machine (SVM) classifier to derive a 1-dimensional "choice axis" from the 20-dimensional data[58]. We performed 5-fold CV to determine the optimal regularization parameter (C) in the SVM. To obtain two additional axes, we applied PCA on the hyperplane orthogonal to the choice axis[59]. Similar to our body-movement decoding analysis (**Fig. 1e**), we repeated the SVM analyses 10 times in different subsampled trials and analyzed the average population trajectory of choice decoding.

Detection of far neurons

We calculated the cosine distance between each mouse neuron and the ARC units. Neurons with a cosine distance smaller than 0.95 were defined as near neurons, whereas the remaining neurons were defined as far neurons. For analysis involving far neurons, we selected 45 sessions containing at least five far neurons. We then built the HRC using these 45 sessions.

Quantification and statistical analysis

All analyses were performed with MATLAB (MathWorks) or Python3. We used two-sided nonparametric statistical tests. The error bars for the means are standard errors or 95% confidence intervals. The confidence intervals for sound and choice decoding were estimated by bootstrap resampling (1,000 iterations)[60]. We used robust linear models (Python *statsmodels*) and evaluated the significance of regression coefficients with t tests (**Fig. 2e, f, S3a**). We used one-way ANOVA (Python *scipy.stats*) followed by Tukey’s HSD post hoc test to compare multiple groups (**Fig. 2g, S3b**). Statistical details are provided in the figure legends, figures, or the Results section.

Data and code availability

The data and codes are on Google Drive (https://drive.google.com/drive/folders/1ekkqqT8vQ17TGG86ZcH0ds7l6wkbKYS0?usp=drive_link) for editors and reviewers, and will be publicly available at Zenodo

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(https://zenodo.org/records/14749313) before publication.

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Author contributions

Y.U. analyzed the data and wrote the paper. H.M. analyzed the data. S.W. collected the animal data. A.F. designed the study, analyzed the data, and wrote the paper.

Declaration of interests

Y.U. and A.F. are inventors on a filed patent application for hybrid-input neural network, submitted to the Japan Patent Office.

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Declaration of interests

☐ The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

☒ The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Akihiro Funamizu, Yutaro Ueoka has patent pending to The University of Tokyo. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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